

Assignment for Advanced Topics in Computer Science

LLM-Assisted Big Data Integration

Work mode	Groups of two students, or individually
Due date	June 15, 2026
Submission	PDF report and GitHub repository

1. Overview

The assignment focuses on designing, implementing, and evaluating a small LLM-assisted Big Data Integration pipeline. The system must integrate multiple heterogeneous, partially overlapping, and possibly conflicting sources describing the same domain.

Students will compare a traditional integration baseline (possibly involving “classical” ML technique) with a pipeline that uses Large Language Models in selected stages of the process. The goal is not to show that LLMs are always superior, but to understand where they help, where they fail, and how they can be combined with classical techniques.

2. Integration Workflow

The pipeline must address the three main phases discussed in class:

1. Schema alignment: map source attributes to a mediated schema.
2. Record linkage / entity resolution: identify records referring to the same real-world entity.
3. Data fusion / truth discovery: reconcile conflicting values and produce one integrated representation per entity.

3. Required Pipelines

Pipeline A - Traditional Baseline

- Schema matching based on names, data types, value overlap, or string similarity.
- Blocking for candidate generation.
- Pairwise record matching based on similarity functions.
- Clustering of matching records.
- Data fusion based on rules, majority voting, recency, or simple source weights.

Pipeline B - LLM-Assisted Integration

The LLM-assisted pipeline must use an LLM in at least two (or more) of the following stages.

Stage	Possible LLM Role	Expected Output
Schema alignment	Map attributes using names, samples, data types, and domain context.	JSON mappings with confidence and possible abstentions.
Record linkage	Judge difficult candidate pairs or verify borderline cases.	Match/non-match decision with confidence and short explanation.
Value normalization	Standardize names, dates, addresses, units, or noisy descriptions.	Canonical value and normalization type.
Data fusion	Select or normalize the most plausible value when sources disagree.	Selected value, confidence, evidence, and abstention if needed.

4. Dataset and Scope

Choose an established public benchmark whenever possible. Suitable domains include products, restaurants, books, movies, companies, universities, bibliographic records, cultural heritage objects, and open government data. Recommended sources include Magellan datasets for entity matching, ER-Magellan and WDC Product Matching datasets used in Ditto as references for neural entity matching, JedAI/Mendeley ER benchmarks for entity resolution, Web Data Commons benchmarks for schema matching and table annotation, and the Alaska

benchmark for product data integration. General portals such as Kaggle, Web Data Commons, data.gov, data.europa.eu, DBLP/OpenAlex/Crossref, Wikidata, and MusicBrainz may also be used, provided that students clearly define the integration task and construct or obtain suitable ground truth..

The dataset should be large enough to make blocking, ML-based matching, and selective LLM invocation meaningful. Minimum requirements:

- 3 heterogeneous sources;
- 1,000 records in total;
- at least 5 relevant attributes in the mediated schema (individual sources may contain only a subset of these attributes, may use different names for them, and may include additional attributes);
- 200 integrated entities;
- 300 known positive matching pairs or equivalent cluster-level ground truth;
- 100 conflicting attribute values requiring fusion.

Larger scalability experiments are welcome, but in this case LLM calls could be applied selectively, for example only to uncertain mappings, borderline pairs, or unresolved conflicts.

If complete ground truth is not available, evaluate on a labeled subset while running the pipeline on the full dataset. The report must clearly distinguish between the full input dataset and the labeled evaluation set. If students construct the dataset themselves, they must document sources, collection process, sampling strategy, and ground-truth construction.

5. Model and Prompting Requirements

- Use at least one LLM, such as GPT, Claude, Gemini, Llama, Qwen, or a comparable local/open model.
- Use structured outputs, preferably JSON.
- Include all prompts in the repository.
- Report model settings, including temperature, maximum tokens, and number of runs.
- Treat invalid JSON, missing fields, hallucinated values, and empty responses as failures unless a documented fallback is used.
- Do not manually correct LLM outputs at evaluation time.

6. Experimental Comparison

At minimum, compare the following configurations:

- Traditional baseline: no LLM is used for integration decisions.
- LLM-assisted pipeline: the LLM is used in at least two integration stages.

Optional comparisons may include LLM-only schema alignment, LLM-only record linkage, LLM-only fusion, zero-shot versus few-shot prompting, small versus larger models, or LLM use only for uncertain cases.

7. Evaluation Metrics

Component	Required Metrics
Schema alignment	Accuracy or precision/recall/F1 on the attribute mappings.
Record linkage	Pairwise precision, recall, and F1 on predicted matches. Also report the number of candidate pairs after blocking.
Data fusion	Accuracy of the selected values on the attributes for which ground truth is available.
End-to-end integration	Overall quality discussion based on the final integrated dataset and on the main errors observed.

Additional metrics such as reduction ratio, completeness, cost, latency, or hallucination rate are optional and may be included when useful.

8. Deliverables

Technical Report

Submit a concise PDF report containing:

- Methodology: dataset, mediated schema, baseline pipeline, LLM-assisted pipeline, prompts, output formats, blocking, matching, clustering, and fusion strategies.
- Experimental protocol: dataset size, sources, ground truth, parameters, thresholds, model settings, prompt versions, number of LLM calls, and failure policy.
- Results: quantitative tables or plots comparing baseline and LLM-assisted configurations.
- Error analysis: at least three concrete examples showing source records, system output, expected output, and explanation of the error.
- Discussion: where LLMs help, where deterministic methods remain preferable, and how cost, latency, hallucinations, and reproducibility affect system design.
- GitHub link.

GitHub Repository

- Executable Python scripts or notebooks;
- input data or download instructions;
- ground-truth files;
- all prompts and model configurations;
- requirements.txt, environment.yml, or equivalent;
- README with reproduction instructions;
- code to compute all reported metrics;
- final integrated dataset or scripts to regenerate it.

9. Grading

Criterion	Weight	Description
Implementation	25%	Correct implementation of baseline and LLM-assisted pipelines.
Metrics	20%	Correct evaluation of schema alignment, linkage, fusion, and end-to-end quality.
Experimental design	20%	Reproducible dataset, prompts, parameters, code, and environment.
Error analysis	20%	Clear discussion of when LLMs help, fail, or introduce new errors.
Presentation and code quality	10%	Readable report, well-organized repository, clean code.
Originality	5%	Advanced extensions or particularly insightful analysis.

10. Optional Extensions

- LLM-assisted meta-blocking or candidate-pair pruning.
- LLM-based source-quality estimation.
- Copy detection and discounting of redundant sources.
- Temporal data fusion for values that evolve over time.
- Human-in-the-loop integration with simulated feedback.
- Prompt sensitivity analysis or cost-aware integration.

11. Practical Tips

- Start with a deterministic baseline before adding the LLM.
- Use blocking aggressively: do not call the LLM on all possible pairs unless the dataset is tiny.
- Keep prompts short, explicit, and versioned.
- Require structured output and log every LLM response.
- Treat hallucinated or unsupported values as errors.
- Evaluate each component separately before evaluating the final dataset.
- The goal is to understand the role of LLMs in Big Data Integration, not only to maximize a score.